

TidyTuesday Week 38

Pranay Gundam

Sunday 20th September, 2026

Contents

1	Weekly Summary	2
2	Date: 2026-09-14	3
2.1	Raw Regression	3
2.2	Detrended Regression	5
3	Date: 2026-09-15	7
3.1	Raw Regression	7
3.2	Detrended Regression	9
4	Date: 2026-09-16	11
4.1	Raw Regression	11
4.2	Detrended Regression	13
5	Date: 2026-09-17	15
5.1	Raw Regression	15
5.2	Detrended Regression	17
6	Date: 2026-09-18	19
6.1	Raw Regression	19
6.2	Detrended Regression	21
7	Date: 2026-09-19	23
7.1	Raw Regression	23
7.2	Detrended Regression	25
8	Date: 2026-09-20	27
8.1	Raw Regression	27
8.2	Detrended Regression	29

1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to the weekly ritual where we let a computer grab two economic time series that have never met, force them into a linear regression, and then act shocked at the results. This week's contestants included California house prices, a discontinued corporate loan survey, carbon dioxide from distillate fuel, Japanese retail stock sentiment, and, in a genuinely inspired pairing, SNAP benefit recipients in Walker County, Alabama versus retail sales in Estonia. As always: these are random pairings, no causality is implied, and the correct emotional response to any "significant" finding here is a raised eyebrow, not a policy memo.

Monday opened with a bang that turned into a whimper. House prices in Oxnard-Thousand Oaks-Ventura versus discontinued financial-business investment posted a jaw-dropping R-squared of 0.87 in the raw regression, the kind of number that makes you want to call a press conference. Then we detrended it and the R-squared face-planted to 0.23. Translation: two things that both went up over several decades will look madly in love until you remind them that everything went up over several decades. Classic trend mirage, exactly the sort of thing this project exists to ridicule. The rest of the week was a graveyard of near-misses – San Luis Obispo house prices versus a loan survey couldn't clear significance in either version (congratulations on the honesty), and the emerging-markets bond index versus piped gas prices had a gorgeous raw fit that collapsed into statistical mush once detrended, its p-value ballooning to a proud 0.32.

The actually interesting cases were the quiet ones that survived the detrending gauntlet. U.S. Assets versus Asian business confidence (Wednesday) stayed significant in both regressions, and the relationship even flipped sign after detrending, which is the kind of plot twist that suggests something real is wobbling around under the shared drift. Thursday's health-care "other separations" versus retail trade hires held up too, and honestly two labor-market flow measures having a genuine link is almost disappointingly plausible – we come here for nonsense, not for things that make sense. And then Saturday's grand finale: SNAP recipients in one Alabama county versus Estonian retail sales stayed significant even after detrending, a correlation so gloriously unhinged that I refuse to theorize about it beyond noting that the universe has a sense of humor.

So the scoreboard: a couple of raw hits that dissolved the instant we asked them to prove themselves, and a handful of relationships that stubbornly refused to die. Do remember that "survived detrending" is a low bar for evidence and not a claim that Alabama food assistance is secretly steering Baltic shopping habits. It's a hobby, the correlations are spurious until proven otherwise, and proving otherwise is emphatically not the goal. See you next week for more randomly arranged marriages of convenience between innocent data series.

2 Date: 2026-09-14

Series ID: ATNHPIUS37100Q

This series is titled All-Transactions House Price Index for Oxnard-Thousand Oaks-Ventura, CA (MSA) and has a frequency of Quarterly. The units are Index 1995:Q1=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1976-04-01 and the observation end date is 2026-04-01. The popularity of this series is 6.

Series ID: FBGFIEQ027S

This series is titled Financial business; gross fixed investment, nonresidential equipment, software, and structures, Flow (DISCONTINUED) and has a frequency of Quarterly. The units are Millions of Dollars and the seasonal adjustment is Seasonally Adjusted Annual Rate. The observation start date is 1946-10-01 and the observation end date is 2017-10-01. The popularity of this series is 1.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_FBGFIEQ027S	R-squared:	0.870
Model:	OLS	Adj. R-squared:	0.869
Method:	Least Squares	F-statistic:	1103.
Date:	Mon, 14 Sep 2026	Prob (F-statistic):	5.84e-75
Time:	17:07:08	Log-Likelihood:	-1926.0
No. Observations:	167	AIC:	3856.
Df Residuals:	165	BIC:	3862.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1.341e+04	3837.676	3.493	0.001	5829.236	2.1e+04
value_fred_ATNHPIUS37100Q	745.7296	22.457	33.207	0.000	701.389	790.070

Omnibus:	1.131	Durbin-Watson:	0.039
Prob(Omnibus):	0.568	Jarque-Bera (JB):	0.803
Skew:	-0.148	Prob(JB):	0.669
Kurtosis:	3.165	Cond. No.	341.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

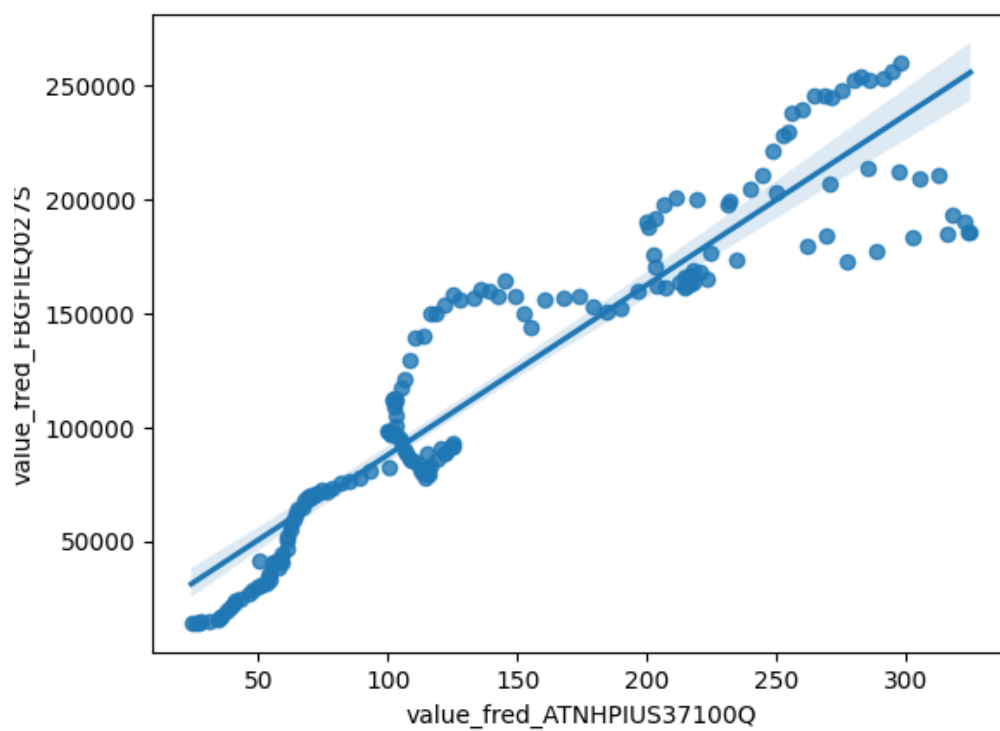


Figure 1: Raw Regression Plot for 2026-09-14

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_FBGFIEQ027S_detrended	R-squared:	0.233
Model:	OLS	Adj. R-squared:	0.228
Method:	Least Squares	F-statistic:	50.05
Date:	Mon, 14 Sep 2026	Prob (F-statistic):	4.06e-11
Time:	17:07:08	Log-Likelihood:	-1814.4
No. Observations:	167	AIC:	3633.
Df Residuals:	165	BIC:	3639.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-1.573e-09	985.135	-1.6e-12	1.000	-1945.095	1945.095
value_fred_ATNHPIUS37100Q_detrended	197.7479	27.952	7.075	0.000	142.559	252.937

Omnibus:	3.074	Durbin-Watson:	0.100
Prob(Omnibus):	0.215	Jarque-Bera (JB):	2.616
Skew:	-0.248	Prob(JB):	0.270
Kurtosis:	3.361	Cond. No.	35.2

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

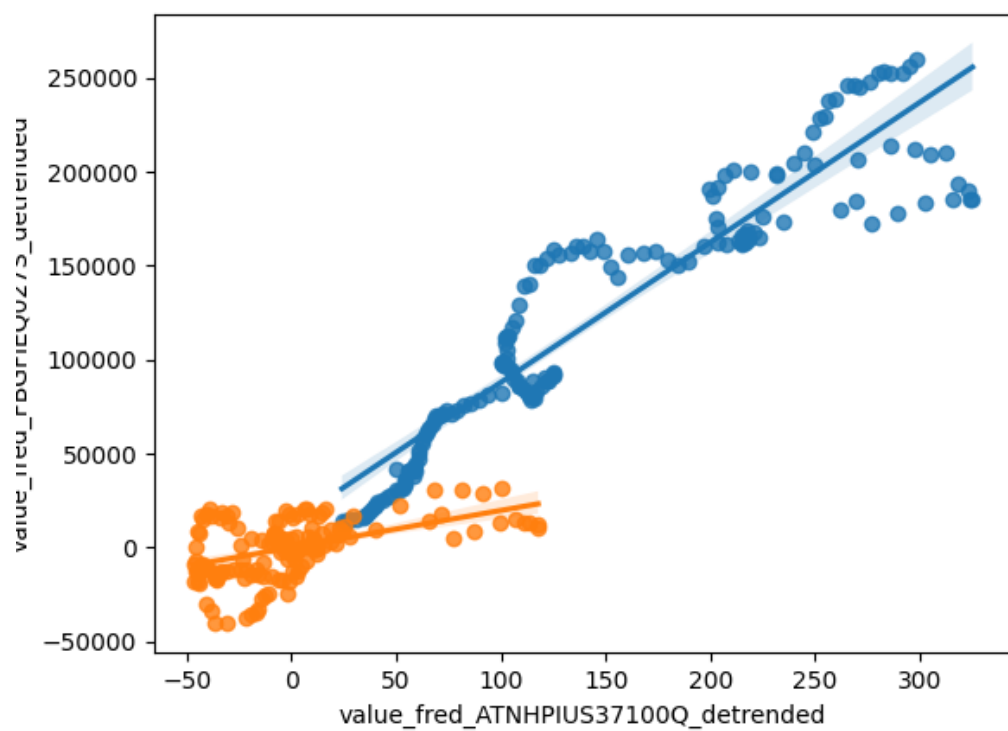


Figure 2: Detrended Regression Plot for 2026-09-14

3 Date: 2026-09-15

Series ID: ATNHPIUS42020Q

This series is titled All-Transactions House Price Index for San Luis Obispo-Paso Robles, CA (MSA) and has a frequency of Quarterly. The units are Index 1995:Q1=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1977-10-01 and the observation end date is 2026-04-01. The popularity of this series is 8.

Series ID: EARPRXDBNQ

This series is titled Percent of Value of Loans Prime Based by Average Loan Size (\$ thousands), Prime, Domestic Banks (DISCONTINUED) and has a frequency of Quarterly, 2nd Month's 1st Full Week. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1997-04-01 and the observation end date is 2017-04-01. The popularity of this series is 1.

3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_EARPRXDBNQ	R-squared:	0.039
Model:	OLS	Adj. R-squared:	0.027
Method:	Least Squares	F-statistic:	3.241
Date:	Tue, 15 Sep 2026	Prob (F-statistic):	0.0756
Time:	15:46:24	Log-Likelihood:	-364.17
No. Observations:	81	AIC:	732.3
Df Residuals:	79	BIC:	737.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	176.5714	8.976	19.671	0.000	158.704	194.438
value_fred_ATNHPIUS42020Q	0.0686	0.038	1.800	0.076	-0.007	0.144

Omnibus:	0.239	Durbin-Watson:	1.271
Prob(Omnibus):	0.887	Jarque-Bera (JB):	0.081
Skew:	-0.077	Prob(JB):	0.960
Kurtosis:	3.017	Cond. No.	866.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

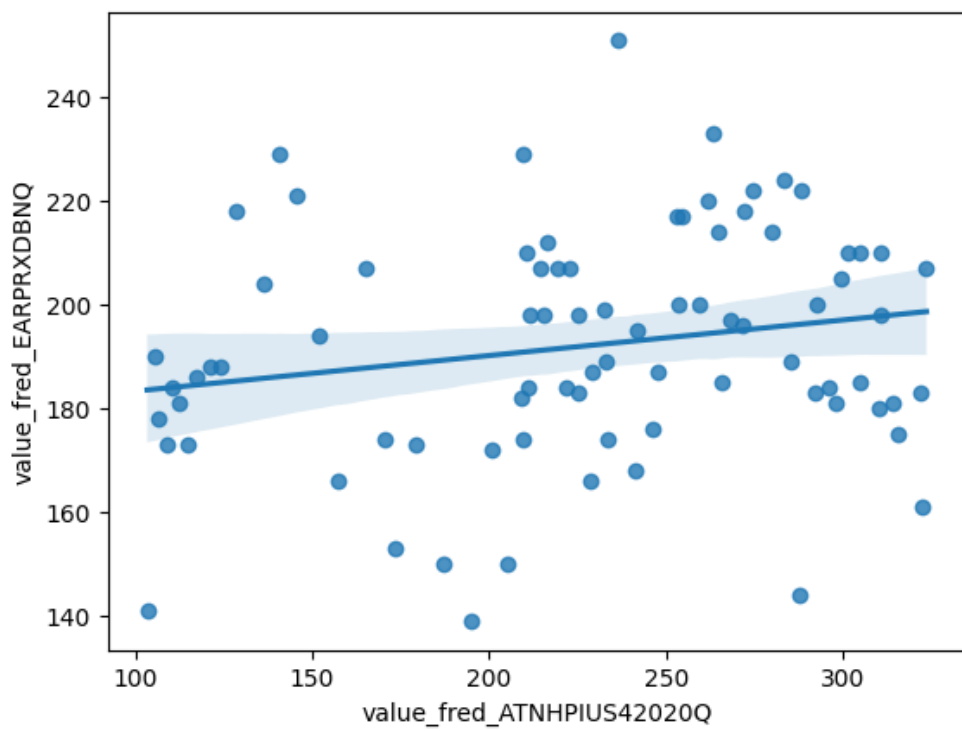


Figure 3: Raw Regression Plot for 2026-09-15

3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_EARPRXDBNQ_detrended	R-squared:	0.027
Model:	OLS	Adj. R-squared:	0.015
Method:	Least Squares	F-statistic:	2.188
Date:	Tue, 15 Sep 2026	Prob (F-statistic):	0.143
Time:	15:46:25	Log-Likelihood:	-356.02
No. Observations:	81	AIC:	716.0
Df Residuals:	79	BIC:	720.8
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-4.581e-12	2.207	-2.08e-12	1.000	-4.393	4.393
value_fred_ATNHPIUS42020Q_detrended	-0.0708	0.048	-1.479	0.143	-0.166	0.024

Omnibus:	1.694	Durbin-Watson:	1.556
Prob(Omnibus):	0.429	Jarque-Bera (JB):	1.096
Skew:	0.241	Prob(JB):	0.578
Kurtosis:	3.303	Cond. No.	46.1

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

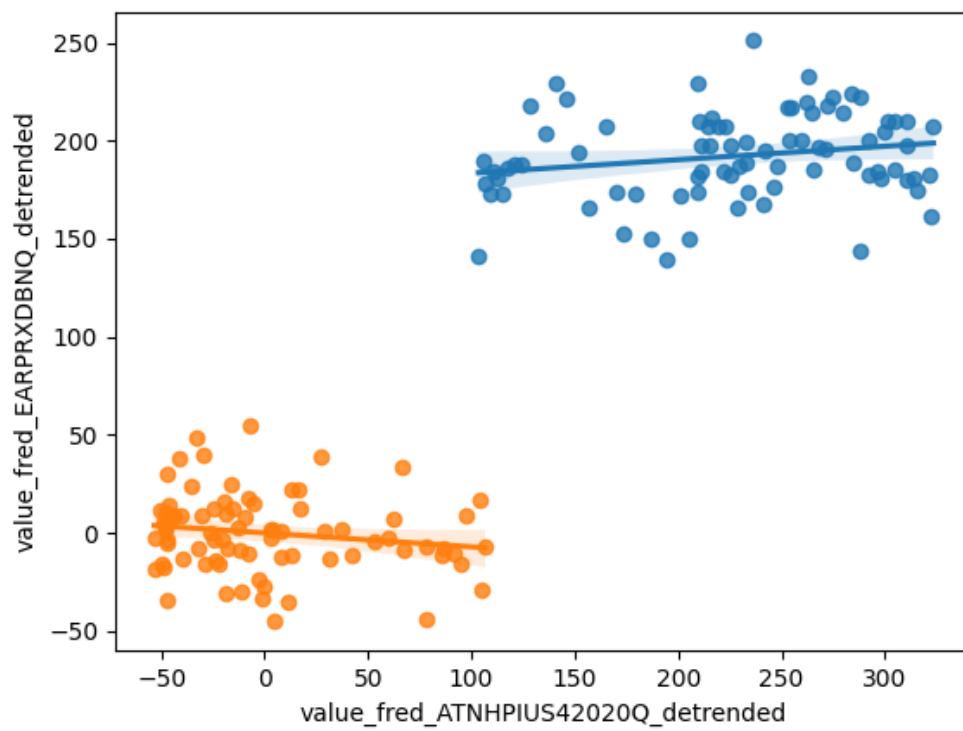


Figure 4: Detrended Regression Plot for 2026-09-15

4 Date: 2026-09-16

Series ID: EMISSCO2VDFCCBA

This series is titled Distillate Fuel Commercial Sector Carbon Dioxide Emissions and has a frequency of Annual. The units are Million Metric Tons CO2 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1973-01-01 and the observation end date is 2024-01-01. The popularity of this series is 1.

Series ID: BRVSLV02JPQ460S

This series is titled Business Tendency Surveys: Volume of Stocks: Economic Activity: Retail Trade, Except of Motor Vehicles and Motorcycles: Current for Japan and has a frequency of Quarterly. The units are Percentage balance and the seasonal adjustment is Seasonally Adjusted. The observation start date is 1990-10-01 and the observation end date is 2026-04-01. The popularity of this series is 1.

4.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_BRVSLV02JPQ460S	R-squared:	0.481
Model:	OLS	Adj. R-squared:	0.465
Method:	Least Squares	F-statistic:	29.65
Date:	Wed, 16 Sep 2026	Prob (F-statistic):	5.44e-06
Time:	15:39:06	Log-Likelihood:	-112.36
No. Observations:	34	AIC:	228.7
Df Residuals:	32	BIC:	231.8
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-20.4615	6.786	-3.015	0.005	-34.285	-6.638
value_fred_EMISSCO2VDFCCBA	1.2106	0.222	5.445	0.000	0.758	1.664

Omnibus:	6.618	Durbin-Watson:	1.027
Prob(Omnibus):	0.037	Jarque-Bera (JB):	5.041
Skew:	-0.838	Prob(JB):	0.0804
Kurtosis:	3.867	Cond. No.	178.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

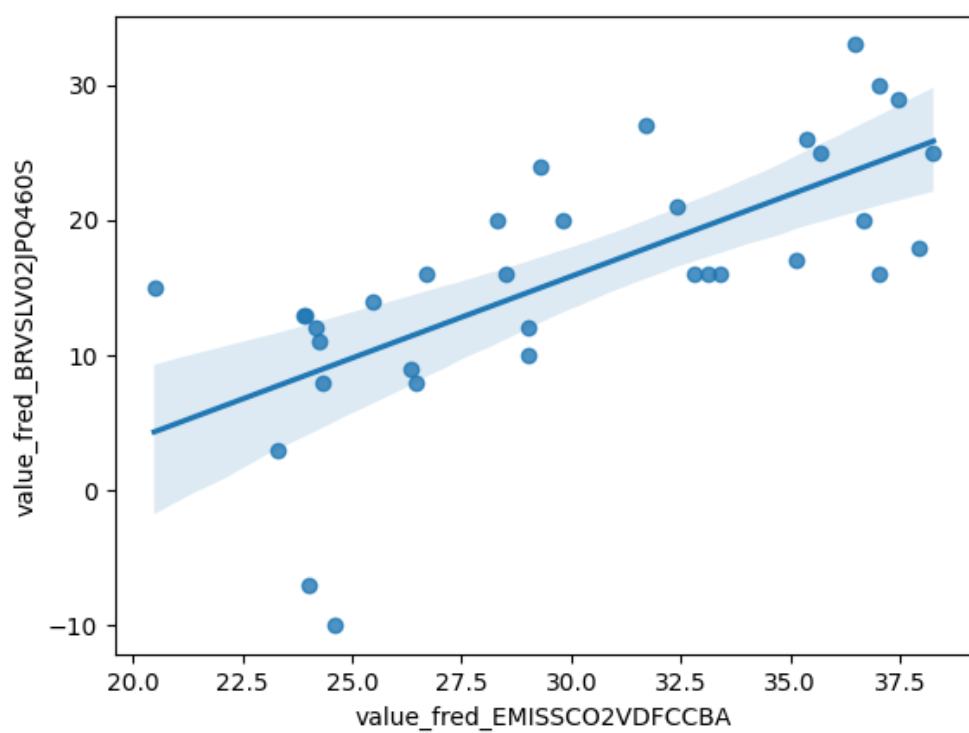


Figure 5: Raw Regression Plot for 2026-09-16

4.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_BRVSLV02JPQ460S_detrended	R-squared:	0.166
Model:	OLS	Adj. R-squared:	0.140
Method:	Least Squares	F-statistic:	6.379
Date:	Wed, 16 Sep 2026	Prob (F-statistic):	0.0167
Time:	15:39:06	Log-Likelihood:	-100.08
No. Observations:	34	AIC:	204.2
Df Residuals:	32	BIC:	207.2
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-6.513e-13	0.812	-8.02e-13	1.000	-1.654	1.654
value_fred_EMISSCO2VDFCCBA_detrended	-1.0600	0.420	-2.526	0.017	-1.915	-0.205

Omnibus:	7.112	Durbin-Watson:	1.687
Prob(Omnibus):	0.029	Jarque-Bera (JB):	5.544
Skew:	-0.868	Prob(JB):	0.0625
Kurtosis:	3.947	Cond. No.	1.93

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

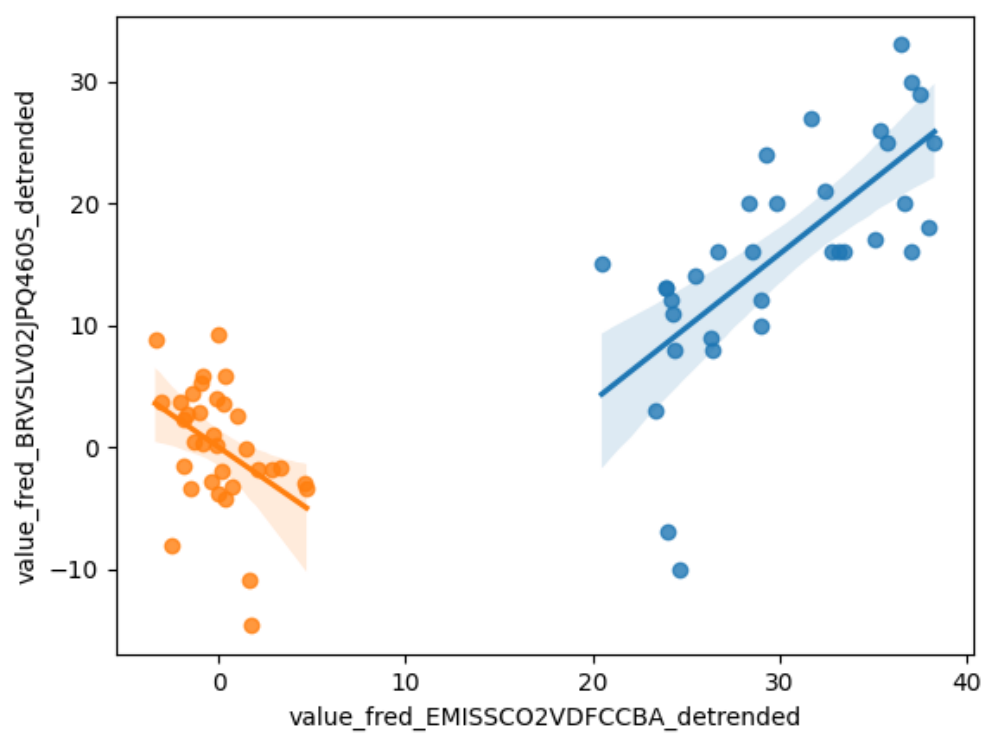


Figure 6: Detrended Regression Plot for 2026-09-16

5 Date: 2026-09-17

Series ID: IIPUSASSQ

This series is titled U.S. Assets and has a frequency of Quarterly, End of Period. The units are Millions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2006-01-01 and the observation end date is 2026-01-01. The popularity of this series is 10.

Series ID: BSCICP035AM665S

This series is titled Composite Leading Indicators: Composite Business Confidence Amplitude Adjusted for Major Five Asia Economies and has a frequency of Monthly. The units are Normalised (Normal=100) and the seasonal adjustment is Seasonally Adjusted. The observation start date is 2000-02-01 and the observation end date is 2023-12-01. The popularity of this series is 1.

5.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_BSCICP035AM665S	R-squared:	0.083
Model:	OLS	Adj. R-squared:	0.070
Method:	Least Squares	F-statistic:	6.363
Date:	Thu, 17 Sep 2026	Prob (F-statistic):	0.0139
Time:	15:45:16	Log-Likelihood:	-123.79
No. Observations:	72	AIC:	251.6
Df Residuals:	70	BIC:	256.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	101.6727	0.802	126.848	0.000	100.074	103.271
value_fred_IIPUSASSQ	-8.045e-08	3.19e-08	-2.522	0.014	-1.44e-07	-1.68e-08

Omnibus:	13.288	Durbin-Watson:	0.449
Prob(Omnibus):	0.001	Jarque-Bera (JB):	16.146
Skew:	-0.826	Prob(JB):	0.000312
Kurtosis:	4.628	Cond. No.	1.25e+08

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.25e+08. This might indicate that there are strong multicollinearity or other numerical problems.

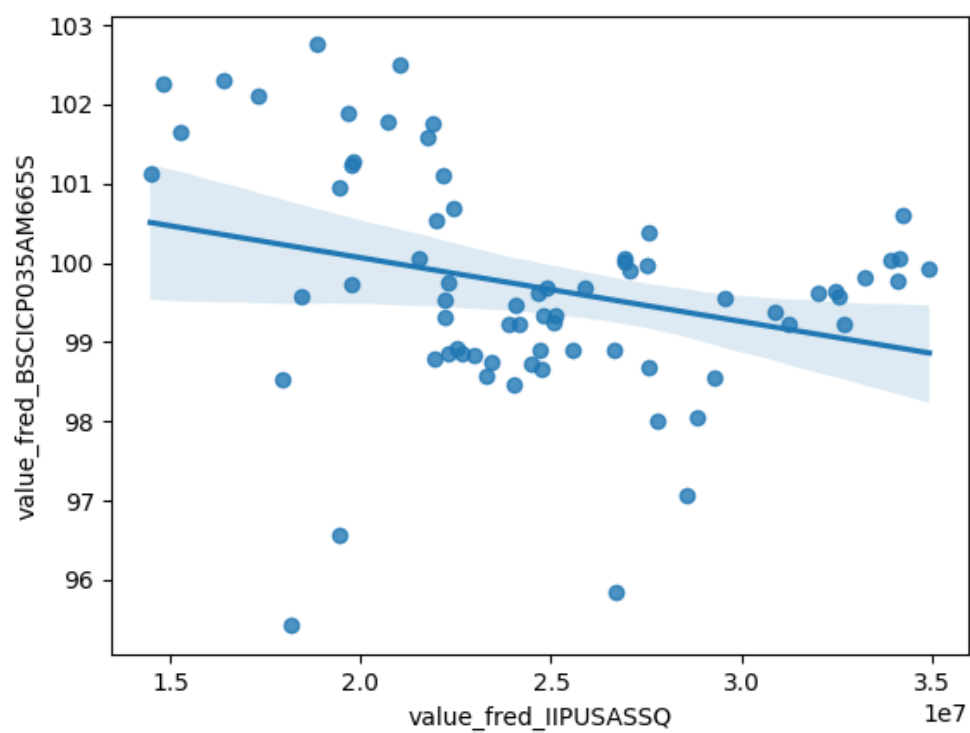


Figure 7: Raw Regression Plot for 2026-09-17

5.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_BSCICP035AM665S_detrended	R-squared:	0.178
Model:	OLS	Adj. R-squared:	0.167
Method:	Least Squares	F-statistic:	15.21
Date:	Thu, 17 Sep 2026	Prob (F-statistic):	0.000219
Time:	15:45:16	Log-Likelihood:	-112.66
No. Observations:	72	AIC:	229.3
Df Residuals:	70	BIC:	233.9
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-2.082e-12	0.138	-1.51e-11	1.000	-0.276	0.276
value_fred_IIPUSASSQ_detrended	3.494e-07	8.96e-08	3.900	0.000	1.71e-07	5.28e-07

Omnibus:	26.580	Durbin-Watson:	0.538
Prob(Omnibus):	0.000	Jarque-Bera (JB):	53.882
Skew:	-1.260	Prob(JB):	1.99e-12
Kurtosis:	6.407	Cond. No.	1.54e+06

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.54e+06. This might indicate that there are strong multicollinearity or other numerical problems.

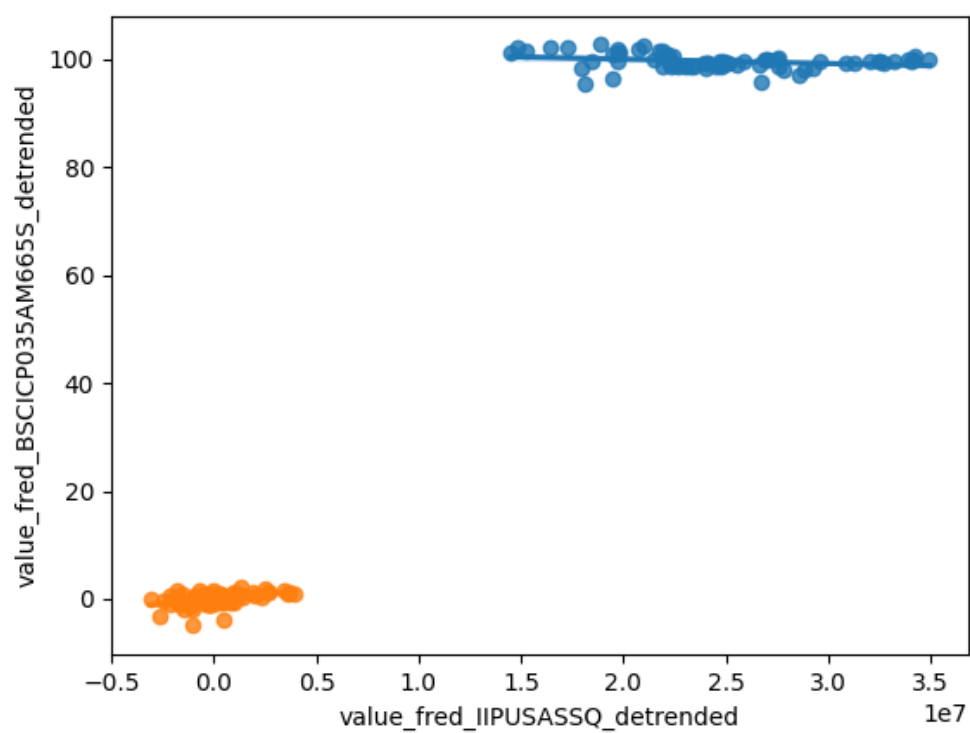


Figure 8: Detrended Regression Plot for 2026-09-17

6 Date: 2026-09-18

Series ID: BAMLEMHGHGLCRPIUSTRIV

This series is titled ICE BofA High Grade US Emerging Markets Liquid Corporate Plus Index Total Return Index Value and has a frequency of Daily. The units are Index and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2023-09-19 and the observation end date is 2026-09-17. The popularity of this series is 2.

Series ID: CUUR0000SEHF02

This series is titled Consumer Price Index for All Urban Consumers: Utility (Piped) Gas Service in U.S. City Average and has a frequency of Monthly. The units are Index 1982-1984=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1935-03-01 and the observation end date is 2026-08-01. The popularity of this series is 9.

6.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_CUUR0000SEHF02	R-squared:	0.732
Model:	OLS	Adj. R-squared:	0.717
Method:	Least Squares	F-statistic:	49.05
Date:	Fri, 18 Sep 2026	Prob (F-statistic):	1.54e-06
Time:	15:17:42	Log-Likelihood:	-71.934
No. Observations:	20	AIC:	147.9
Df Residuals:	18	BIC:	149.9
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-12.4578	36.807	-0.338	0.739	-89.787	64.871
value_fred_BAMLEMHGHGLCRPIUSTRIV	1.0247	0.146	7.004	0.000	0.717	1.332

Omnibus:	1.439	Durbin-Watson:	0.633
Prob(Omnibus):	0.487	Jarque-Bera (JB):	0.769
Skew:	-0.480	Prob(JB):	0.681
Kurtosis:	2.969	Cond. No.	4.45e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 4.45e+03. This might indicate that there are strong multicollinearity or other numerical problems.

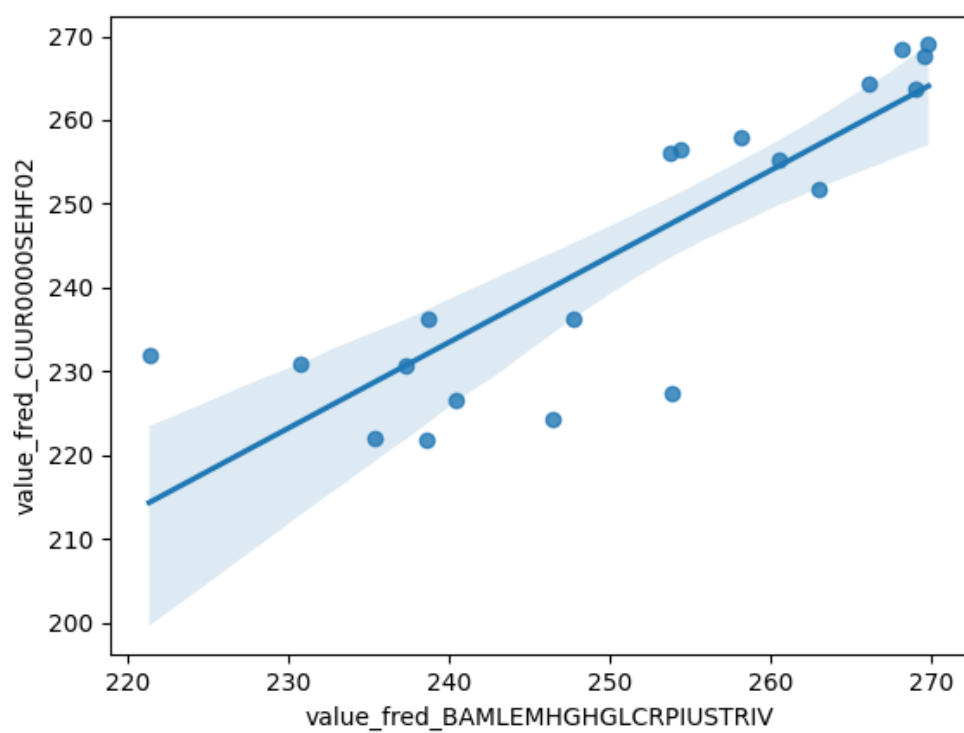


Figure 9: Raw Regression Plot for 2026-09-18

6.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_CUUR0000SEHF02_detrended	R-squared:	0.056
Model:	OLS	Adj. R-squared:	0.003
Method:	Least Squares	F-statistic:	1.063
Date:	Fri, 18 Sep 2026	Prob (F-statistic):	0.316
Time:	15:17:42	Log-Likelihood:	-66.774
No. Observations:	20	AIC:	137.5
Df Residuals:	18	BIC:	139.5
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1.553e-11	1.607	9.66e-12	1.000	-3.377	3.377
value_fred_BAMLEMHGHGLCRPIUSTRIV_detrended	-0.4521	0.438	-1.031	0.316	-1.373	0.469

Omnibus:	3.477	Durbin-Watson:	0.858
Prob(Omnibus):	0.176	Jarque-Bera (JB):	1.332
Skew:	0.090	Prob(JB):	0.514
Kurtosis:	1.749	Cond. No.	3.67

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

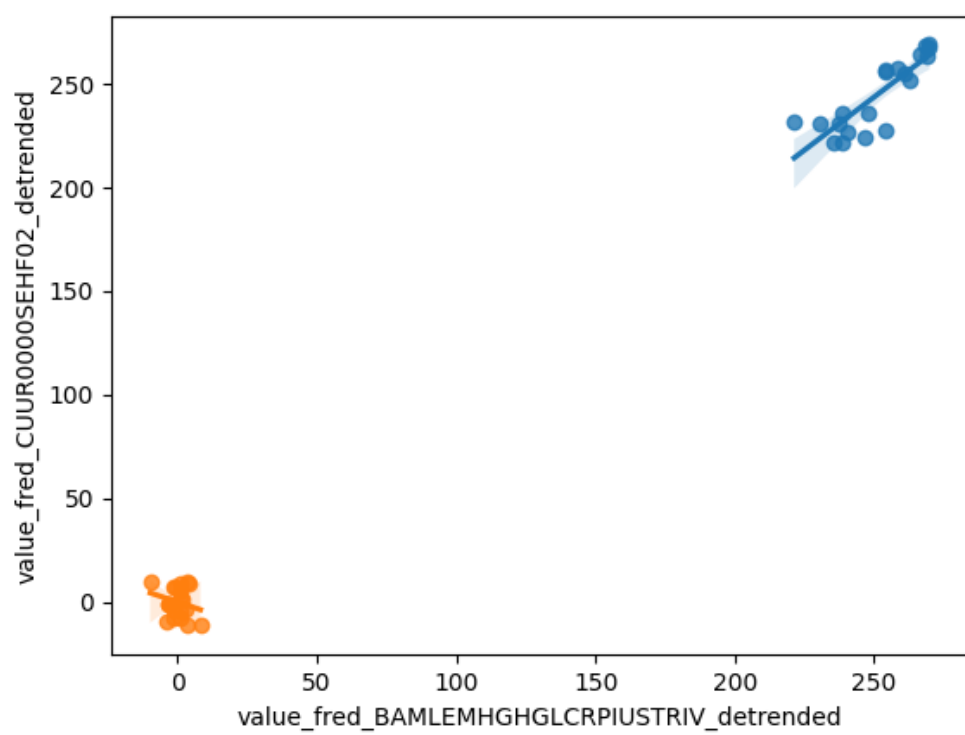


Figure 10: Detrended Regression Plot for 2026-09-18

7 Date: 2026-09-19

Series ID: JTU6200OSR

This series is titled Other Separations: Health Care and Social Assistance and has a frequency of Monthly. The units are Rate and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2000-12-01 and the observation end date is 2026-07-01. The popularity of this series is 1.

Series ID: JTU4400HIR

This series is titled Hires: Retail Trade and has a frequency of Monthly. The units are Rate and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2000-12-01 and the observation end date is 2026-07-01. The popularity of this series is 1.

7.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_JTU4400HIR	R-squared:	0.011			
Model:	OLS	Adj. R-squared:	0.007			
Method:	Least Squares	F-statistic:	3.313			
Date:	Sat, 19 Sep 2026	Prob (F-statistic):	0.0697			
Time:	14:54:03	Log-Likelihood:	-414.91			
No. Observations:	308	AIC:	833.8			
Df Residuals:	306	BIC:	841.3			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	4.9928	0.175	28.499	0.000	4.648	5.338
value_fred_JTU6200OSR	-1.5143	0.832	-1.820	0.070	-3.151	0.123
Omnibus:	5.215	Durbin-Watson:	0.754			
Prob(Omnibus):	0.074	Jarque-Bera (JB):	5.214			
Skew:	0.318	Prob(JB):	0.0737			
Kurtosis:	2.971	Cond. No.	16.3			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

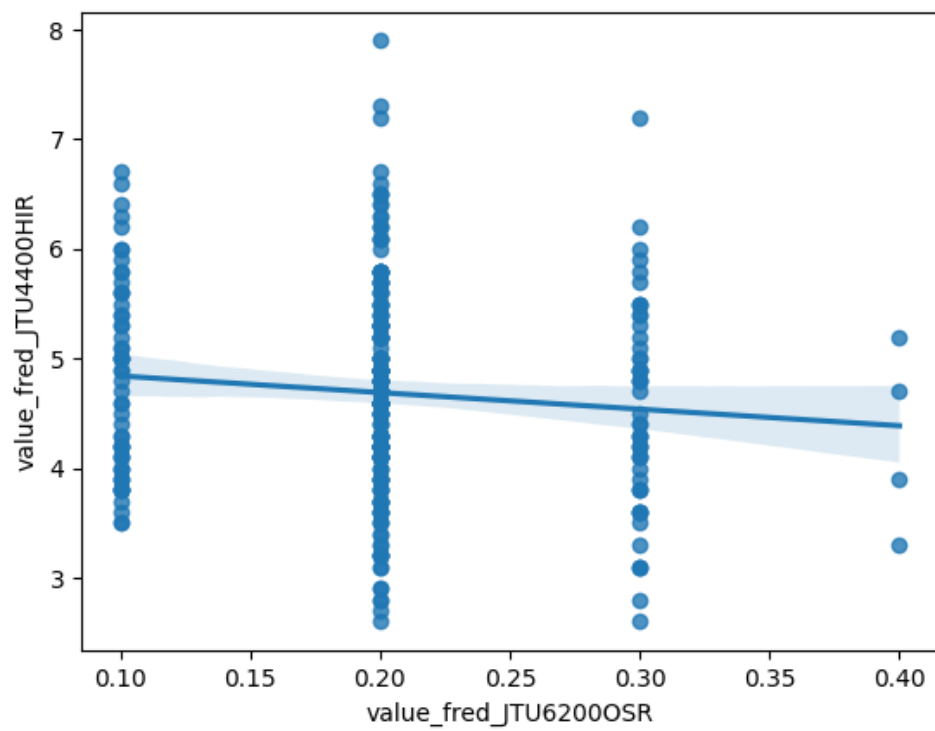


Figure 11: Raw Regression Plot for 2026-09-19

7.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_JTU4400HIR_detrended	R-squared:	0.023
Model:	OLS	Adj. R-squared:	0.020
Method:	Least Squares	F-statistic:	7.277
Date:	Sat, 19 Sep 2026	Prob (F-statistic):	0.00737
Time:	14:54:03	Log-Likelihood:	-409.05
No. Observations:	308	AIC:	822.1
Df Residuals:	306	BIC:	829.6
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	7.77e-13	0.052	1.49e-11	1.000	-0.103	0.103
value_fred_JTU6200OSR_detrended	-2.2825	0.846	-2.698	0.007	-3.947	-0.618

Omnibus:	7.615	Durbin-Watson:	0.790
Prob(Omnibus):	0.022	Jarque-Bera (JB):	7.440
Skew:	0.352	Prob(JB):	0.0242
Kurtosis:	3.289	Cond. No.	16.2

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

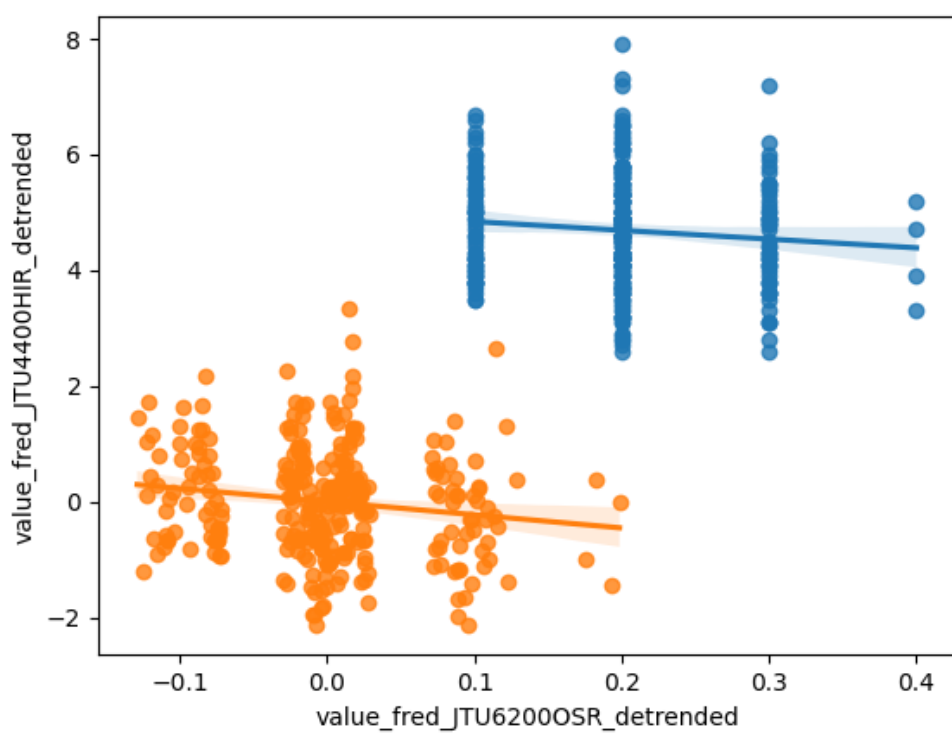


Figure 12: Detrended Regression Plot for 2026-09-19

8 Date: 2026-09-20

Series ID: CBR01127ALA647NCEN

This series is titled SNAP Benefits Recipients in Walker County, AL and has a frequency of Annual. The units are Persons and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1989-01-01 and the observation end date is 2023-01-01. The popularity of this series is 1.

Series ID: SLRTO02EEA661S

This series is titled Sales: Retail Trade: Total Retail Trade: Value for Estonia and has a frequency of Annual. The units are Index 2015=100 and the seasonal adjustment is Seasonally Adjusted. The observation start date is 1995-01-01 and the observation end date is 2022-01-01. The popularity of this series is 1.

8.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_SLRTO02EEA661S	R-squared:	0.504
Model:	OLS	Adj. R-squared:	0.484
Method:	Least Squares	F-statistic:	25.41
Date:	Sun, 20 Sep 2026	Prob (F-statistic):	3.36e-05
Time:	14:58:36	Log-Likelihood:	-128.52
No. Observations:	27	AIC:	261.0
Df Residuals:	25	BIC:	263.6
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-1.7917	16.658	-0.108	0.915	-36.100	32.517
value_fred_CBR01127ALA647NCEN	0.0090	0.002	5.040	0.000	0.005	0.013

Omnibus:	7.952	Durbin-Watson:	0.179
Prob(Omnibus):	0.019	Jarque-Bera (JB):	6.195
Skew:	1.112	Prob(JB):	0.0452
Kurtosis:	3.748	Cond. No.	2.74e+04

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 2.74e+04. This might indicate that there are strong multicollinearity or other numerical problems.

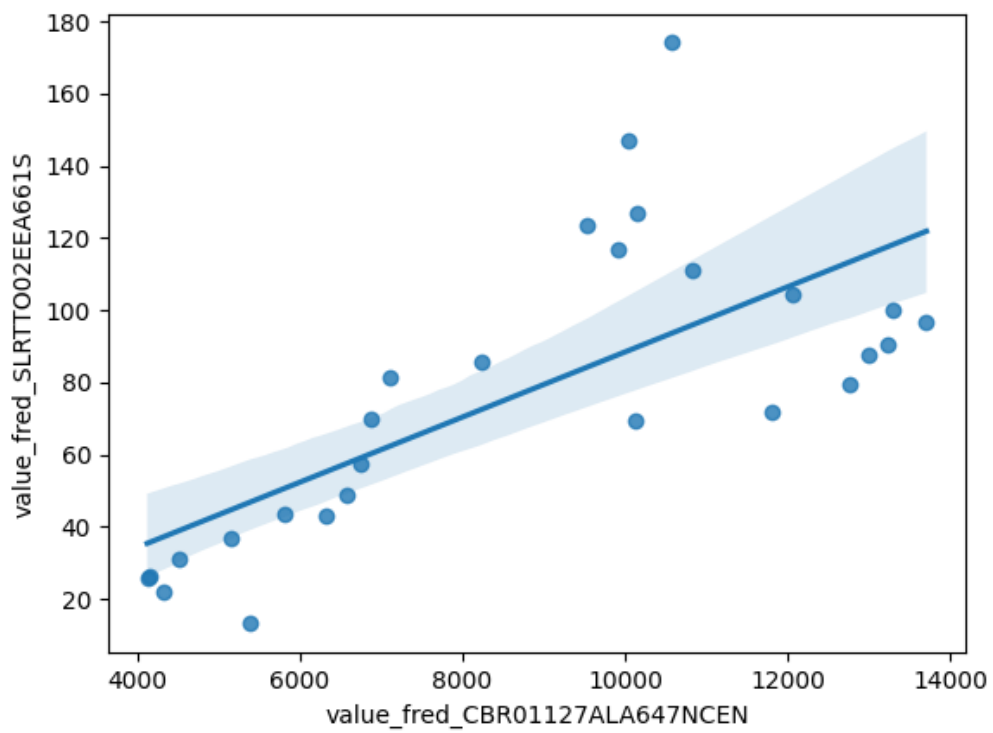


Figure 13: Raw Regression Plot for 2026-09-20

8.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_SLRTTO02EEA661S_detrended	R-squared:	0.247
Model:	OLS	Adj. R-squared:	0.217
Method:	Least Squares	F-statistic:	8.214
Date:	Sun, 20 Sep 2026	Prob (F-statistic):	0.00831
Time:	14:58:36	Log-Likelihood:	-93.880
No. Observations:	27	AIC:	191.8
Df Residuals:	25	BIC:	194.4
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1.344e-12	1.566	8.58e-13	1.000	-3.226	3.226
value_fred_CBR01127ALA647NCEN_detrended	-0.0024	0.001	-2.866	0.008	-0.004	-0.001

Omnibus:	15.687	Durbin-Watson:	0.784
Prob(Omnibus):	0.000	Jarque-Bera (JB):	17.422
Skew:	1.448	Prob(JB):	0.000165
Kurtosis:	5.664	Cond. No.	1.90e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.9e+03. This might indicate that there are strong multicollinearity or other numerical problems.

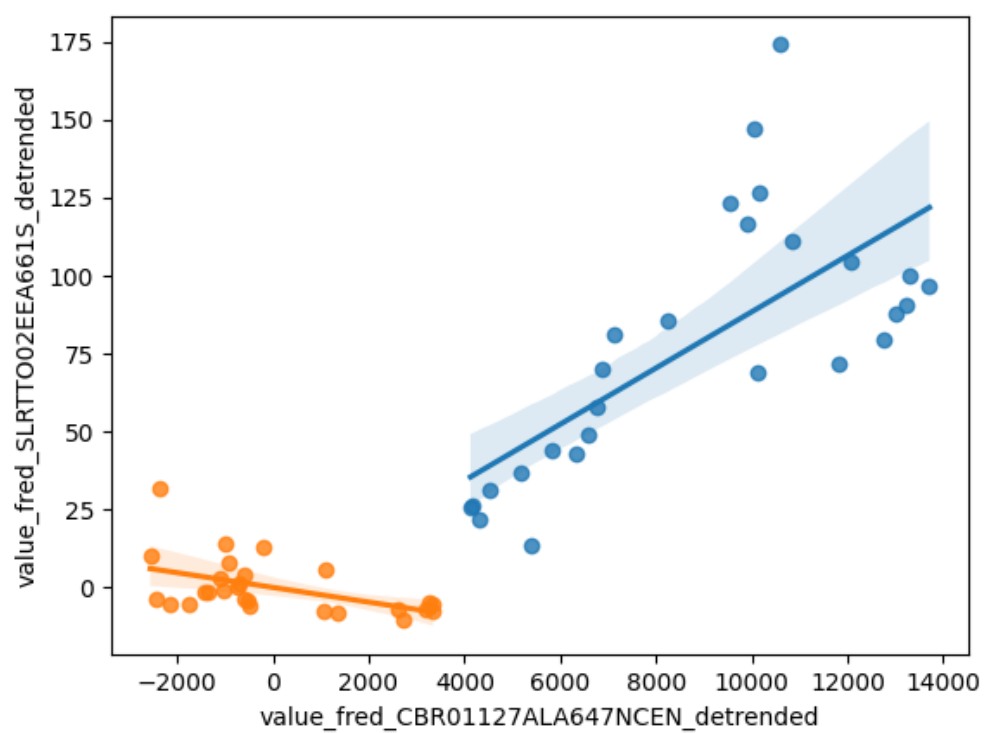


Figure 14: Detrended Regression Plot for 2026-09-20